

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Major areas of AI include Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Robotics, and Expert Systems.

Machine Learning enables computers to learn from data and make predictions without being explicitly programmed.

Deep Learning uses neural networks with many layers to solve complex problems such as image recognition and language translation.

Applications of AI include virtual assistants, recommendation systems, autonomous vehicles, healthcare diagnostics, fraud detection, and smart manufacturing.

AI also raises important ethical issues such as privacy, bias, transparency, and responsible use of intelligent systems.